# Year 7 Mathematics

## Unit 2: Number Skills

2.6 Calculations

# Calculations

STARTER · 5 MIN

**Date:** _____ **Title: Calculations**

**LEARNING OBJECTIVES**

- To be able to estimate answers to complex calculations

**STARTER** *Answer in your book. No calculators.*

1. Round 47 to the nearest 10
2. Round 3.6 to the nearest whole number
3. Work out 20 × 30
4. Work out 600 ÷ 5

# Starter — Answers

STARTER

*Tick or correct in green pen*

1. **50**
2. **4**
3. **600**
4. **120**

# Success Criteria

SUCCESS CRITERIA

*By the end of this lesson:*

| | | |
|---|---|---|
| BRONZE | I can round numbers appropriately | ☐ |
| SILVER | I can use rounded numbers to estimate calculations | ☐ |
| GOLD | I can estimate calculations involving powers and roots and check answers | ☐ |

*Tick each box in the plenary when you can do it on your own.*

# Introduction — Calculations

INTRODUCTION · 3 MIN

An estimate is a sensible guess of the answer, often by rounding first.

## 1 significant figure

Round to the first non-zero digit.

19.6 → 20  347 → 300  0.38 → 0.4

## ≈

Means 'is approximately equal to'.

31.2 ≈ 30

## Estimate

Round every number, then work it out.

4.9 × 3.1 ≈ 5 × 3 = 15

## Checking

Compare your answer with the estimate.

Far off? Check again.

# Examples 1 and 2 — Rounding to 1 Significant Figure

EXAMPLES · 5 MIN

BRONZE

*Watch and copy into your book*

**Hint** 1 s.f.: first non-zero digit  Next digit 5 or more: round up

## Example 1

Round to 1 significant figure:

19.6,  31.2,  5.1

## Example 2

Round to 1 significant figure:

347,  0.38

# Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

*Check your copy and correct in green pen*

## Example 1

**Look at the first digit, then the next**

19.6 = 20

31.2 = 30

**5.1 = 5**

Tip: the next digit 5 or more: round up.

## Example 2

**Keep the place value**

347 = 300

**0.38 = 0.4**

Tip: 347 is not 3: fill with zeros.

# Task 1

PRACTICE · 9 MIN

BRONZE

*Round each number to 1 significant figure*

**Hint**
1 s.f.: first non-zero digit
Next digit 5 or more: round up

1. 68
2. 432
3. 7.8
4. 0.53
5. 2 961
★. 0.0072

# Task 1 — Answers

CORRECTIONS

BRONZE

*Tick or correct in green pen*

1. **70**
2. **400**
3. **8**
4. **0.5**
5. **3000**
- ★ **0.007**

# Examples 3 and 4 — Estimating Calculations

EXAMPLES · 5 MIN

SILVER

*Watch and copy into your book*

**Hint** Round every number first  Show the rounded calculation

## Example 3

Estimate

$4.9 \times 21.3$

## Example 4

Estimate

$(38.7 \times 9.2) \div 2.1$

# Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

*Check your copy and correct in green pen*

## Example 3

**Round each number to 1 s.f.**

$4.9 \approx 5$ and $21.3 \approx 20$

Estimate $= 5 \times 20$

$\mathbf{= 100}$

Tip: write the rounded calculation.

## Example 4

**Round, then work it out**

$38.7 \approx 40$, $9.2 \approx 9$, $2.1 \approx 2$

Estimate $= (40 \times 9) \div 2$

$= 360 \div 2$

$\mathbf{= 180}$

Tip: brackets first.

# Task 2

PRACTICE · 9 MIN

SILVER

*Estimate: show the rounded calculation*

**Hint**
Round every number first
Show the rounded calculation

1. $6.8 \times 3.1$
2. $48 \times 21$
3. $597 \div 3.2$
4. $(9.8 \times 41) \div 1.9$
5. $(29.4 + 11.2) \times 4.8$

★ $(61 \times 4.2) \div 0.48$

# Task 2 — Answers

CORRECTIONS

SILVER

*Tick or correct in green pen*

1. $7 \times 3 = 21$
2. $50 \times 20 = 1000$
3. $600 \div 3 = 200$
4. $(10 \times 40) \div 2 = 200$
5. $(30 + 10) \times 5 = 200$

★ $(60 \times 4) \div 0.5 = 480$

# Examples 5 and 6 — Estimating with Powers and Roots

EXAMPLES · 5 MIN

GOLD

*Exam-style: show all your working*

**Hint** Use square numbers you know
Estimate, then compare

## Example 5

(2 marks)

Estimate

$$\sqrt{48} \times 3.1^2$$

## Example 6

(2 marks)

Sam says $21.3 \times 4.8 = 1\,022.4$

Use an estimate to check.

# Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

*Check your copy and correct in green pen*

## Example 5

**Use square numbers you know**

$\sqrt{48} \approx \sqrt{49} = 7$

$3.1^2 \approx 3^2 = 9$

Estimate $= 7 \times 9$

$= \mathbf{63}$

Tip: 48 is close to the square number 49.

## Example 6

**Estimate, then compare**

Estimate $= 20 \times 5$

$= 100$

**1 022.4 is far too big: Sam is wrong.**

Tip: the real answer is 102.24.

# Task 3

PRACTICE · 9 MIN

GOLD

*Estimate: show the rounded calculation*

**Hint** Use square numbers you know
Estimate, then compare

1. $\sqrt{99}$

2. $4.1^2$

3. $\sqrt{63} + 2.9^2$

4. $5.9^2 \times 2.1$

5. Is $19.8 \times 5.1 = 1\,009.8$ sensible? Explain.

★ $\sqrt{8.9 \times 4.1}$

# Task 3 — Answers

CORRECTIONS

GOLD

*Tick or correct in green pen*

1 $\sqrt{100} = 10$

2 $4^2 = 16$

3 $8 + 9 = 17$

4 $36 \times 2 = 72$

5 No: $20 \times 5 = 100$

★ $\sqrt{9 \times 4} = \sqrt{36} = 6$

# Plenary — Exit Ticket

PLENARY · 5 MIN

*Answer in your book. Try all three.*

**BRONZE** Round 7.48 to 1 significant figure.

**SILVER** Estimate $39 \times 5.2$

**GOLD** Estimate $\sqrt{26} \times 4$

**Reflect:** Look back at the learning objectives. Which ones can you do now?

# Plenary — Answers

PLENARY

*Tick or correct in green pen*

BRONZE **7**

SILVER **40 × 5 = 200**

GOLD **5 × 4 = 20**

# Exam Practice — Rounding

EXAM PRACTICE · 10 MIN

BRONZE

*IGCSE exam-style: show all your working*

**Hint**
1 s.f.: first non-zero digit
Next digit 5 or more: round up

## Question 1

(2 marks)

Round to 1 significant figure.

**(a)** 734 (1)

.....

**(b)** 0.68 (1)

.....

## Question 2

(2 marks)

Round each number to 1 significant figure to estimate 8.7 × 3.2

.....

# Exam Practice — Estimating

EXAM PRACTICE · 10 MIN

SILVER

*IGCSE exam-style: show all your working*

**Hint**
Round every number first
Show the rounded calculation

**Question 3** (2 marks)

Estimate the value of
$(29.6 \times 41.2) \div 6.1$

Show the rounded calculation you use.

.....

**Question 4** (2 marks)

Estimate the value of
$512 \div 9.8$

.....

# Exam Practice — Estimating and Checking

EXAM PRACTICE · 10 MIN

GOLD

*IGCSE exam-style: show all your working*

**Hint** Use square numbers you know
Estimate, then compare

## Question 5

(2 marks)

Estimate the value of

$\sqrt{80} \times 2.9^2$

.....

## Question 6

(2 marks)

Lina says $38.2 \times 5.1 = 1\,948.2$

Use estimation to show that she is wrong.

.....

# Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 1

**(a) 700** B1

**(b) 0.7** B1

Tip: keep the place value.

## Question 2

**9 × 3 = 27** M1 A1

Tip: show the rounded numbers.

# Exam Practice — Mark Scheme

CORRECTIONS

SILVER

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 3

$(30 \times 40) \div 6 \quad = 1200 \div 6$ **M1**

$\mathbf{= 200}$ **A1**

Tip: round every number to 1 s.f.

## Question 4

**$500 \div 10 \quad = 50$** **M1 A1**

Tip: $9.8 \approx 10$.

# Exam Practice — Mark Scheme

CORRECTIONS

GOLD

*Mark your work: M = method mark, A = accuracy mark, B = independent mark*

## Question 5

$\sqrt{80} \approx \sqrt{81} = 9$ M1

**$9 \times 3^2 = 81$** A1

Tip: 80 is close to 81 = $9^2$.

## Question 6

$40 \times 5 = 200$ M1

**So 1 948.2 is far too big** A1

Tip: the real answer is 194.82.